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Techno-economic Analysis and Life Cycle Assessment of CO₂ Electrochemical Conversion via SOEC for Sustainable Industrial Applications

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Abstract

Electrochemical conversion of carbon dioxide (CO₂) to carbon monoxide (CO) via solid oxide electrolysis cells (SOECs) presents a promising pathway for reducing greenhouse gas emissions while producing industrially valuable products. CO is favored over other electrochemical products due to its market demand, favorable economics, and role in synthetic fuel production. This study develops and validates a mechanistic SOEC model using Aspen Custom Modeler (ACM) to quantify the system's performance under varying operational conditions. The model incorporates Butler–Volmer kinetics and thermodynamic formulations to estimate voltage requirements, power consumption, and CO₂ conversion efficiency. Under optimal conditions—500 kW power input and a 98% CO₂ feed composition—the model achieved 82% CO₂ conversion. Validation against experimental data demonstrated consistent trends with respect to voltage, flow rate, temperature, and feed composition. Sensitivity analysis identified power input and operating temperature as the most influential factors affecting system performance, while feed concentration and temperature showed the greatest impact on power consumption. A techno-economic analysis (TEA) estimated capital and operating costs of \$573 and \$134 per tonne of CO, respectively. Complementary life cycle assessment (LCA) revealed the potential for significant CO₂ emission reductions, particularly when powered by renewable electricity sources. These findings underscore the viability of SOEC-based CO₂ electrolysis as a scalable solution for sustainable industrial applications.

Introduction

Overview of Electrolysis in Industrial CO₂ Utilization

In industrial processes, carbon dioxide (CO₂) is commonly emitted as a byproduct and can be captured using methods such as direct air capture or point-source carbon capture technologies [2]. Once captured, CO₂ can serve as a feedstock for producing value-added chemicals through electrochemical conversion. One

important route involves reducing CO_2 to carbon monoxide (CO), which, when combined with hydrogen (H_2), forms synthesis gas (syngas) for downstream fuel and chemical production. An overview of this integrated CO_2 utilization pathway is shown in Fig. 1.

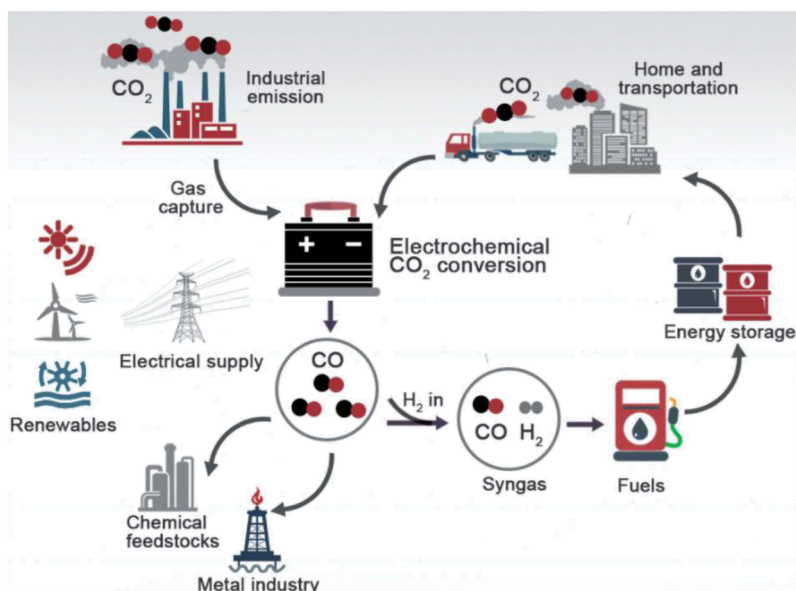


Figure 1—Overview of electrolysis-based CO_2 conversion in industrial applications [2].

Electrochemical CO_2 Reduction Pathways

Electrolysis cells facilitate non-spontaneous reactions through the application of electrical energy. A typical electrolysis cell comprises an anode, a cathode, and an electrolyte. During CO_2 electrolysis, CO_2 is reduced at the cathode to form CO, as illustrated in Fig. 2.

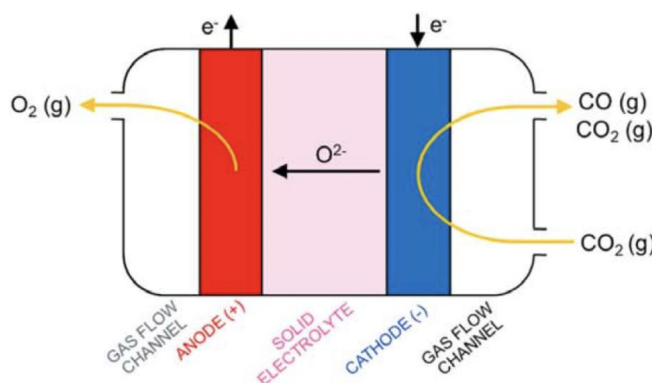


Figure 2—Schematic diagram of CO_2 electrochemical reduction in a generic electrolysis cell [3].

While various products can be synthesized from CO_2 —such as methane, ethylene, and alcohols—the focus of this work is exclusively on the electrochemical production of CO due to its industrial demand and favorable reaction characteristics. The product distribution is strongly influenced by the catalyst material. For example, catalysts such as nickel alloys or silver (Ag) favor CO formation, whereas tin oxide (SnO_2) tends to promote the formation of formate or formic acid derivatives [4].

Types of Electrolysis Cells for CO₂ Conversion

Three primary types of electrolysis cells are employed for CO₂ reduction: molten carbonate electrolysis cells (MCEC), low-temperature aqueous electrolysis cells, and solid oxide electrolysis cells (SOEC), as shown in Fig. 3.

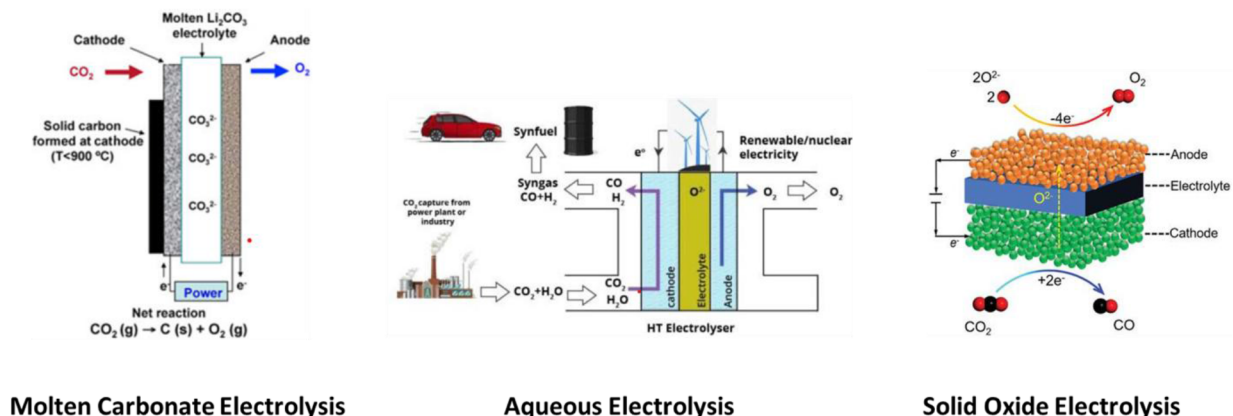


Figure 3—Types of electrolysis cells for CO₂ conversion: Left to right: (a) molten carbonate, (b) aqueous, and (c) solid oxide [5–7].

These technologies differ primarily in their electrolyte materials and operating temperatures. MCECs use molten salts such as Li₂O/Li₂CO₃ and operate at high temperatures with titanium cathodes and graphite anodes. Despite offering advantages such as minimal gas permeation through the electrolyte, MCECs are limited by their low technology readiness level (TRL) and remain at the lab-scale development stage [3].

Aqueous electrolysis cells function at lower temperatures and employ liquid electrolytes. However, they typically suffer from low CO selectivity due to concurrent hydrogen evolution, which complicates product separation [3].

In contrast, SOECs operate at elevated temperatures with solid ceramic electrolytes and offer high selectivity toward CO production. Given their advanced TRL, greater energy efficiency, and product purity, SOECs represent the most promising platform for commercial-scale CO₂-to-CO electrochemical conversion.

Objectives and Scope

Research Motivation and Objectives

Global carbon dioxide (CO₂) emissions continue to rise, with approximately 1.4 trillion metric tons emitted between 1965 and 2017 [1]. To address the dual challenge of reducing emissions while meeting industrial energy demands, there is growing interest in electrochemical pathways for CO₂ utilization. The International Energy Agency (IEA) outlines a net-zero roadmap in which carbon savings must be achieved through current technologies, behavioral changes, and the deployment of new innovations. As shown in Fig. 4, the largest emission reductions in the coming decades must stem from emerging technologies, underscoring the urgency of advancing scalable CO₂ conversion solutions.

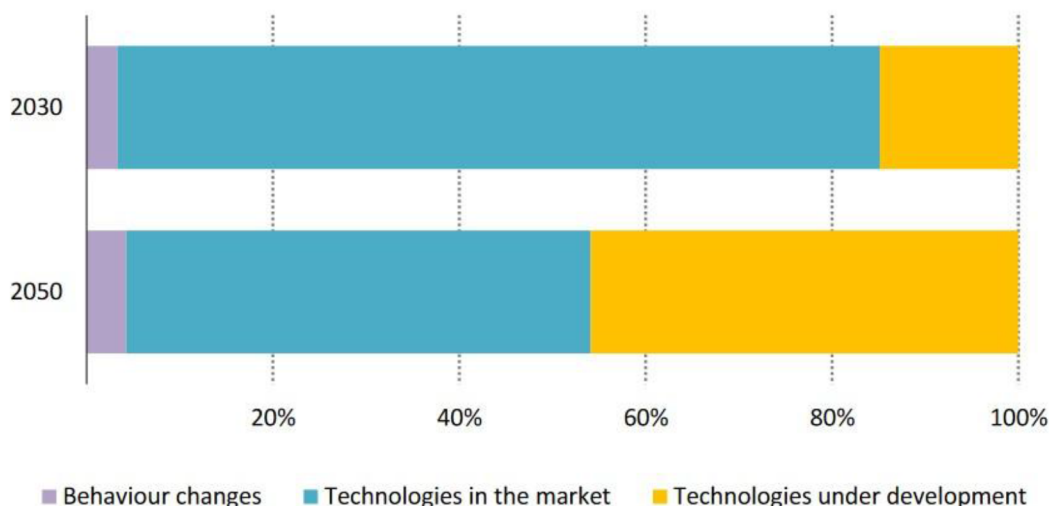


Figure 4—Projected annual CO₂ emissions reductions by source along the net-zero emissions pathway [8].

The goal of this study is to develop and validate a simulation model for a solid oxide electrolysis cell (SOEC) capable of converting CO₂ to CO under industrially relevant conditions. The work involves the identification of essential system parameters and performance outputs, the construction of a robust and explainable electrochemical model, and the integration of the model into a process simulation environment. In addition to the core modeling effort, the study conducts a techno-economic analysis (TEA) to evaluate the capital and operational costs of the system and a life cycle assessment (LCA) to quantify its environmental impact.

Scope and Justification for CO Production

Among the various products that can be derived from electrochemical CO₂ reduction, carbon monoxide (CO) offers the strongest case for near-term commercial deployment. Compared to formic acid—the only other product currently considered economically viable—CO commands approximately 250 times higher demand, with global consumption exceeding 150 million tonnes per year [9]. In addition, CO benefits from a favorable electrochemical profile, with a high molecular weight-to-electron ratio and a relatively low equilibrium potential, making it one of the most energy-efficient CO₂ reduction products [10].

CO is a critical feedstock in syngas-based processes, including Fischer–Tropsch synthesis for synthetic fuels and carbonylation routes for acetic acid production. Its role in producing high-purity metals and chemicals also supports a wide range of industrial applications. Importantly, electrochemical CO production offers an opportunity to decentralize the CO supply chain, enabling small-scale, on-site generation that avoids the risks and costs associated with CO transport and storage [12,13]. The process also eliminates intermediate steps associated with conventional methods, such as CO₂ hydrogenation, and reduces fossil fuel consumption. A wide selection of catalysts—including Au, Ag, Zn, and Ni—has been shown to provide high selectivity toward CO production in electrochemical systems [14–17].

Solid Oxide Electrolysis Cells (SOECs)

Solid oxide electrolysis cells represent one of the most advanced and promising platforms for high-temperature CO₂ electrolysis. Operating at temperatures up to 700 °C and featuring ceramic electrolytes such as nickel-based cermet, SOECs offer high current densities and product selectivity. With a technology readiness level (TRL) of 8 out of 9, SOECs are nearing commercial deployment, with existing pilot demonstrations achieving syngas production capacities up to 7,500 ft³/h [3,18,19].

The electrochemical reactions involved in SOEC operation are as follows:





CO₂ is introduced at the cathode, where it is electrochemically reduced to CO. The resulting oxygen ions migrate through the solid electrolyte to the anode, where oxygen gas is released.

SOECs offer several advantages compared to other electrolyzer types. Their operation in the absence of aqueous media ensures a high-purity CO product, free from hydrogen or water byproducts, reducing downstream separation requirements [3]. High temperatures enhance ionic conductivity and reaction efficiency, resulting in superior current densities, as shown in Fig. 5 [3,21].

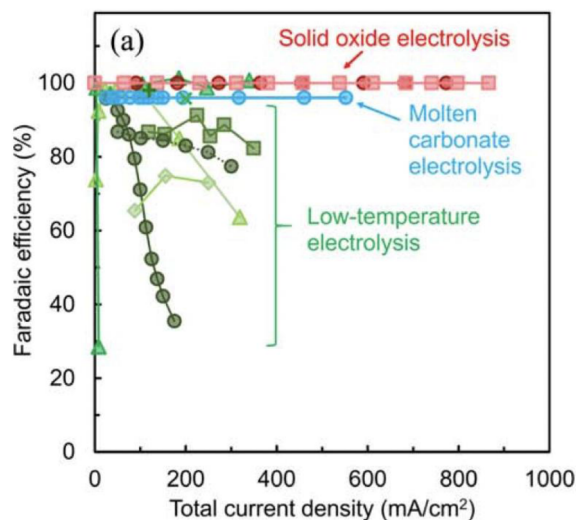


Figure 5—Comparison of SOEC efficiency and current density with other electrolyzer types [3,21].

Furthermore, SOECs exhibit extended operational lifespans, often exceeding one year, due to the chemical stability and corrosion resistance of ceramic materials. In contrast, molten carbonate cells typically degrade within 100 to 120 hours [3,22]. Nonetheless, challenges remain in terms of thermal management, integration into industrial systems, and scalability. Continued research into these areas is critical to realizing the full potential of SOECs for sustainable CO₂ utilization [23].

Methodology

Modeling Framework

Model formulation for the SOEC fundamental equations was carried out in Aspen Custom Modeler (ACM). ACM provides capabilities for writing engineering equations to build customized equipment models and integrate them into Aspen Plus and Aspen HYSYS flowsheets. The modeling workflow followed a structured approach beginning with the identification of key inputs, outputs, and fixed parameters necessary for simulating the electrochemical reduction of CO₂ to CO in a solid oxide electrolysis cell (SOEC). The LCA and TEA analysis were conducted with material balances and cost estimation features in Aspen Plus. This framework is illustrated in Fig. 6.

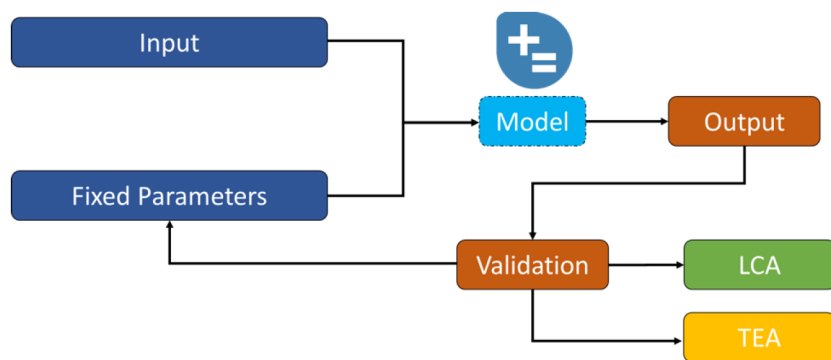


Figure 6—Action plan for the electrochemical reduction model.

Model Inputs and Parameters

The model inputs included applied voltage or current, operating temperature, pressure, and feedstream flowrate and composition. These parameters directly affect the energy consumption and the extent of CO₂ conversion in the system. Specifically, applied voltage influences the rate of electrochemical reactions, while temperature and pressure impact the reaction kinetics and gas-phase behavior. The feed composition and flowrate determine the reactant availability and product distribution.

In addition to operating inputs, the model included several fixed design parameters such as electrode material, electrolyte composition, and catalyst selection. These parameters can be modified within the model but were held constant during baseline simulations. The primary outputs of the model were the production rates of CO and O₂, the amount of unconverted CO₂, and the current density across the electrolyzer.

There are several design parameters associated with the electrolyzer's configuration such as the electrode type, electrolyte composition, and catalysts selection. Those are fixed parameters in the model and can be modified by the user. The main purpose of the model is to simulate the electrochemical reduction of CO₂ to CO. Therefore, the main outputs of the model are the CO and O₂ production, the amount of unreacted CO₂, and the current density of the electrolyzer.

Model Formulation

To model a Solid Oxide Electrolysis Cell (SOEC), there are certain general simplifications and assumptions to facilitate the analysis [24]. These include: (1) assuming steady-state operation of the SOEC; (2) treating gases as ideal; (3) considering a uniform and constant operating temperature throughout the SOEC [25]; (4) presuming complete electrochemical reactions; (5) idealizing the solid electrolyte to assume that no electric current leaks through it; (6) assuming that species concentrations at the electrode interface equal those in the bulk channel flow.

The electric potential is the driving force to initiate the electrochemical reaction in the SOEC. The electrochemical model that was developed consists of the calculation of the Nernst potential, ohmic overpotential, activation overpotential, and concentration overpotential. All of which they are used to characterize the voltage requirement as expressed in Equation 4.

$$V = E + \eta_{ohmic} + \eta_{act,c} + \eta_{act,a} + \eta_{conc,c} + \eta_{conc,a} \quad (4)$$

Where V is the applied voltage, E is the Nernst potential, and η_i are the ohmic, cathode activation, anode activation, cathode concentration, and anode concentration overpotentials, respectively [26]. The Nernst potential refers to reversible potential that predicts the minimum required electrical potential to split CO₂ and it is expressed in Equations 5 & 6 [27] [26].

$$E = E_0 + \frac{RT}{2F} \ln \left[\frac{p_{CO}^0 (p_{O_2}^0)^{\frac{1}{2}}}{p_{CO_2}^0} \right] \quad (5)$$

$$E_0 = \frac{[G_{CO}(T, P) + 0.5G_{O_2}(T, P) - G_{CO_2}(T, P)]}{2F} \quad (6)$$

Where F is Faraday constant, R is the ideal gas constant, T is the temperature, p_i^o is the species i partial pressure at the inlet, and $G_i(T, P)$ is the species Gibbs free energy. A call function in ACM was used to provide the Gibbs free energy for a specified temperature, pressure for each of the pure components.

The property method used for the modelling of the SOEC was Soave-Redlich-Kwong (SRK) equation of state method. The SRK method is a cubic equation of state and used to calculate the standard Gibbs Free Energy. This was then part of the Nernst potential calculation to determine the system voltage. The property method was able to more accurately predict the Gibbs Free Energy due to the cohesion function (7,8). This temperature-dependent cohesion function was able to count for the high temperature behavior observed in the SOEC [28]. ω is the acentric factor and then T_r is the reference temperature of the SOEC.

$$\alpha(T_r, \omega) = [1 + m(1 - T_r^{0.5})]^2 \quad (7)$$

$$\text{where, } m = 0.480 + 1.57 * \omega - 0.176 * \omega^2 \quad (8)$$

The ohmic overpotential accounts for the resistance to the conduction of ions through the electrolyte and electrons through the electrode, interconnects and contact resistances. In this model, only the oxide anion through the electrolyte was considered and it was assumed the remaining resistances are negligible. This is consistent with earlier studies by [29]. The ohmic overpotential based on Ohm's law is expressed in Equation 9 [30].

$$\eta_{ohmic} = J \times L \times 2.99 \times 10^{-5} \exp\left(\frac{10,300}{T}\right) \quad (9)$$

Where J is the current density, and L is the thickness of the electrolyte.

The activation overpotential represents the voltage losses associated with the activation of the adsorbed species, activation of the electrode material and activation energy requires for the charge transfer [27]. The relationship between the electrode activation potential and the current density is defined using Butler-Volmer equation (Equation 10) [31]. The activation overpotential for both cathode and anode is shown in Equation 11.

$$J = J_0 \left[\exp\left(\frac{\alpha z F \eta_{act}}{RT}\right) - \exp\left(-\frac{(1-\alpha) z F \eta_{act}}{RT}\right) \right] \quad (10)$$

$$\eta_{act,i} = \frac{RT}{zF} \sinh^{-1}\left(\frac{J}{2J_{0,i}}\right), i = a \text{ or } c \quad (11)$$

Where J_0 is the exchange current density, α is the symmetrical factor, and z is the number of electrons produced per reaction.

The resistance to the diffusion of reactant and product species is captured in the concentration overpotential. The cathode and anode concentration over potential is expressed in Equations 12 & 13 [30].

$$\eta_{conc,c} = \frac{RT}{2F} \ln \left(\frac{1 + \frac{RTJd_c}{2FD_{CO_2}^{eff} p_{CO}^0}}{1 - \frac{RTJd_c}{2FD_{CO_2}^{eff} p_{CO_2}^0}} \right) \quad (12)$$

$$\eta_{conc,a} = \frac{RT}{4F} \ln \left(\frac{\sqrt{\frac{JRT\mu d_a}{2FB_g} + (p_{O_2}^0)^2}}{p_{O_2}^0} \right) \quad (13)$$

Where d_c is the cathode thickness, d_a is the anode thickness, D_i^{eff} is the effective diffusivity of species i , B_g is the permeability, and μ is the dynamic viscosity. The dynamic viscosity was obtained using a call function in ACM at a given temperature and pressure.

The production flowrate of CO was calculated using Faraday's law in Equation 14. The electric power relation to the cell voltage and current density is represented in Equation 15.

$$\dot{n}_{CO} = \frac{JN_{cell}A_{cell}}{2F} \quad (14)$$

$$P = JA_{cell}N_{cell}V_{cell} \quad (15)$$

Where N_{cell} is the number of cells in SOEC stack, and A_{cell} is the cell active area.

Electrolyzer Design Parameters

To align with commercially relevant systems, the model assumed a cell area of $30 \times 30 \text{ cm}^2$ [32]. The CO production capacity was set to 115 kg/h, consistent with the range of current commercial systems, which operate between 7 and 230 kg CO/h [33].

Results & Discussion

Model Validation

The developed SOEC model in Aspen Custom Modeler was validated against experimental data available in the literature [34] to ensure its robustness and physical accuracy. The validation process involved aligning the model's design specifications with those reported in the benchmark study, including operating temperature, feed composition, and flowrate. While some model parameters were not explicitly provided in the experimental reference, the validation focused on replicating observed trends under varying operating conditions.

Cell voltage predictions were compared to experimental data for three scenarios: variations in CO concentration in the feed (10–50 mol%), operating temperature (700–900 °C), and total feed flowrate (1–5 L/h). Results using both the built-in Gibbs free energy function in the simulation environment and an empirical equation were evaluated. In all cases, the empirical method produced voltage values that more closely matched experimental results.

Figure 7 shows that increasing the CO fraction in the feed leads to an increase in open cell voltage. This is consistent with a reduction in the partial pressure of CO_2 , which increases the Nernst potential. The deviation in voltage prediction compared to experimental data was 17% using the Gibbs function and 10% using the empirical equation.

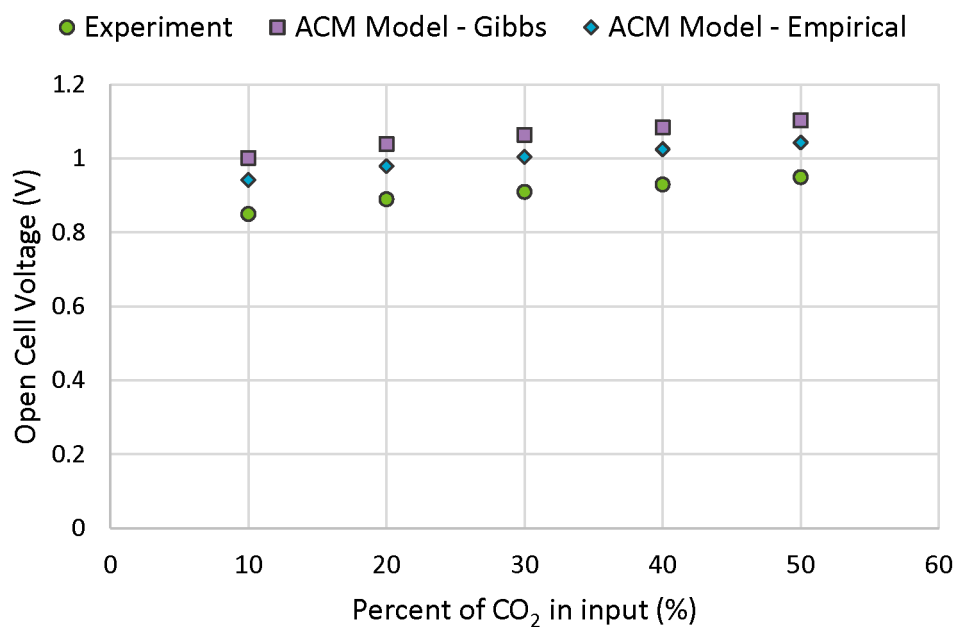


Figure 7—Comparison of experimental and ACM model-predicted cell voltage with varying feed CO₂ concentration.

As shown in Fig. 8, increasing temperature results in a lower open cell voltage due to enhanced reaction kinetics. The deviation from experimental data for this scenario was 17% and 11% for the Gibbs and empirical models, respectively. Figure 9 illustrates that variation in flowrate had negligible influence on cell voltage, consistent across both experimental and simulation results, with respective deviations of 19% and 13%.

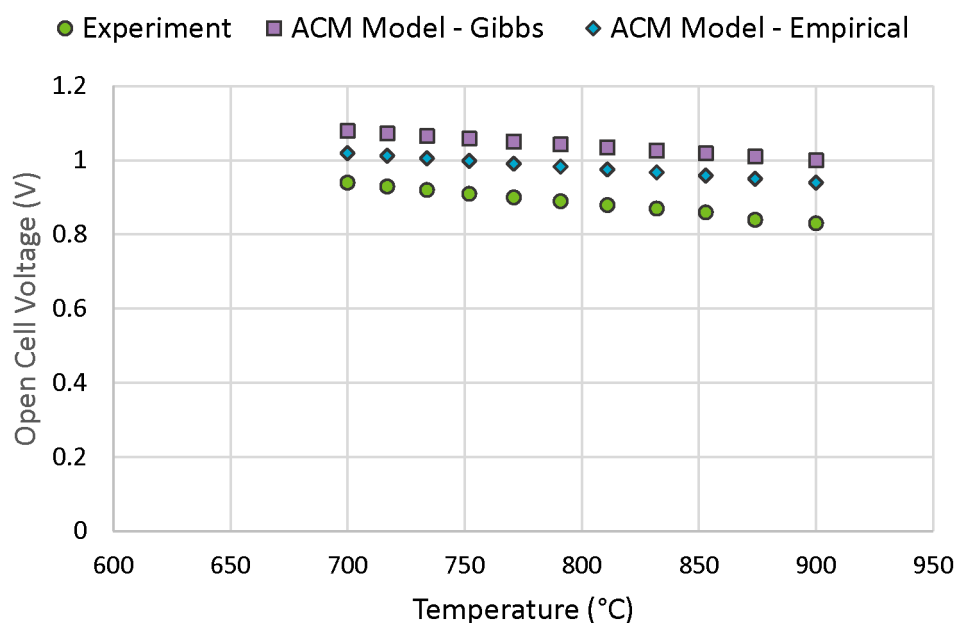


Figure 8—Effect of operating temperature on cell voltage: experimental vs. model.

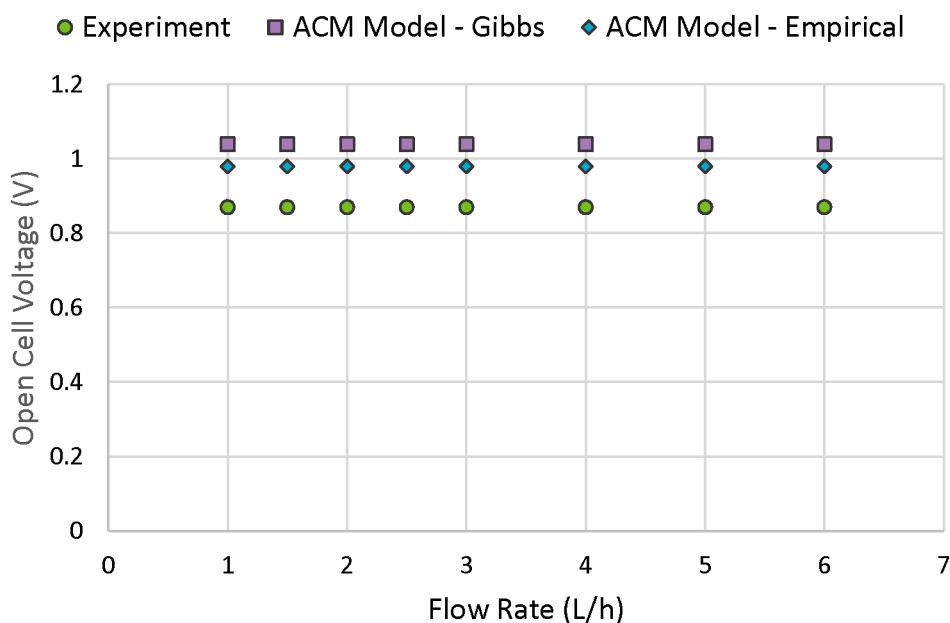


Figure 9—Cell voltage as a function of feed flowrate: experimental vs. model.

Sensitivity Analysis

Following model validation, a sensitivity analysis was conducted to assess the effect of key operating and design parameters on SOEC performance. Table 1 summarizes the base-case input values and corresponding model outputs.

Table 1—Base case input and output values for key model variables.

Input Variables/Parameters	Base Case Value	Output Variables	Base Case Value
Temperature [°C]	700	CO ₂ Conversion [%]	84.1
Power [kW]	300	CO Production [kmol/h]	4.38
Flowrate [kmol/h]	5	Power/CO [kWh/kg]	0.01
CO ₂ Feed Mol [%]	65%	Cell Voltage [V]	2.05
Number of Cells	115	Current Density [A/cm ²]	1.42
Cell Size [m ²]	0.09		
Pressure [atm]	1		

Each parameter was varied by $\pm 50\%$ from the base case to identify the most influential variables. The results, displayed in Figs. 10 and 11 using tornado charts, showed that temperature and power input were the most critical parameters affecting both CO₂ conversion and specific energy consumption. The influence of temperature and power was capped at approximately $\pm 25\%$ in conversion due to the already high baseline conversion (84%).

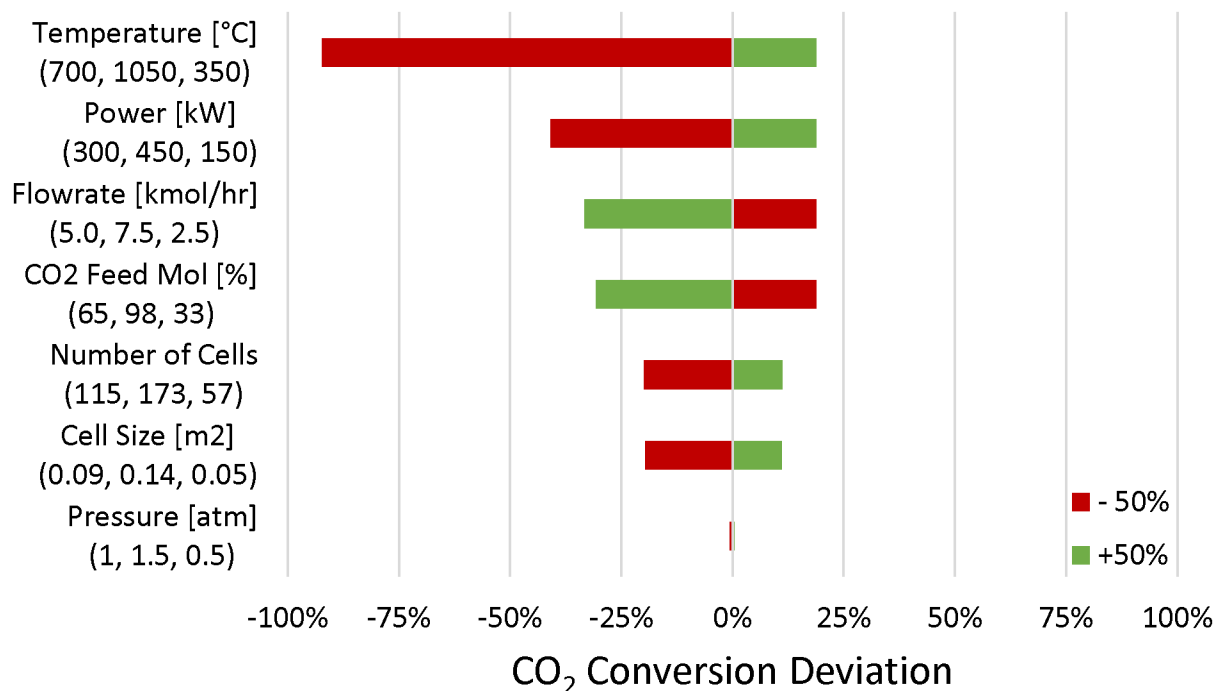


Figure 10—Sensitivity of CO₂ conversion to ±50% variation in key parameters.

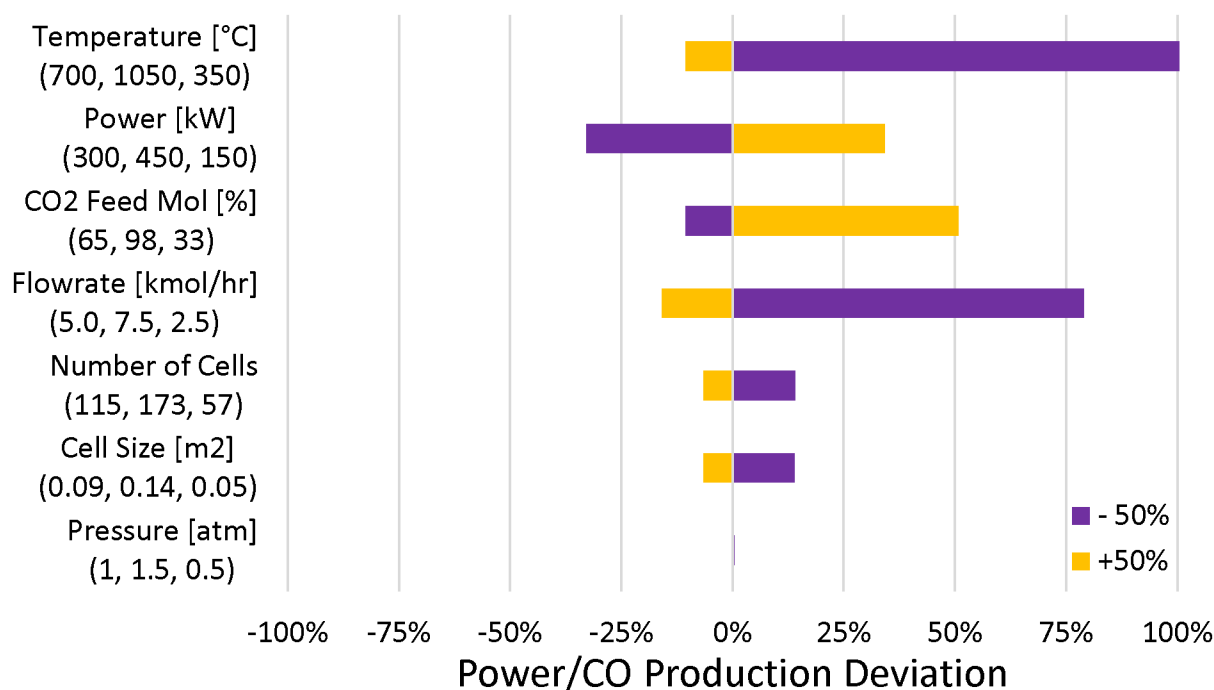


Figure 11—Sensitivity of specific power consumption to ±50% variation in key parameters.

To further investigate the interactive effects of temperature and power, a multivariable analysis was conducted. Figures 12 and 13 illustrate the influence of operating temperature (500–1050 °C) and applied power (50–1000 kW) on CO₂ conversion and power consumption per unit CO production, respectively.

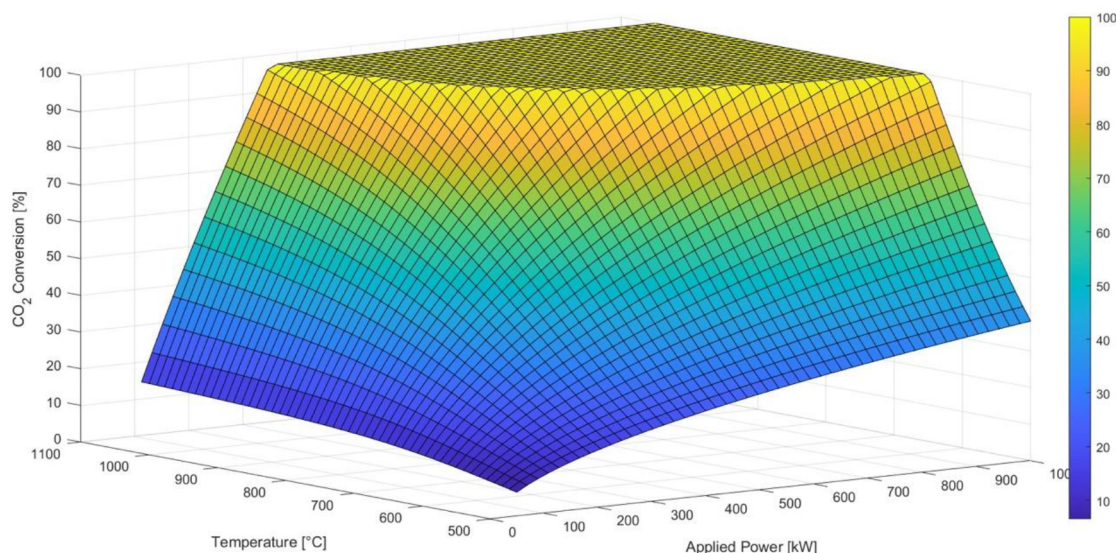


Figure 12—Combined effect of temperature and applied power on CO₂ conversion.

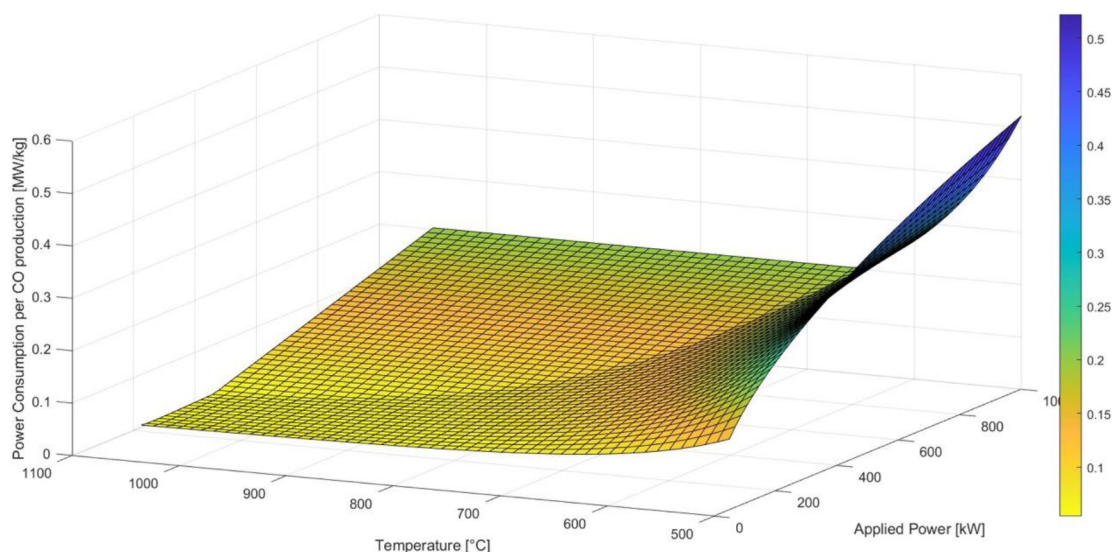


Figure 13—Combined effect of temperature and applied power on specific power consumption for CO production.

As shown in Fig. 12, full conversion can be achieved at specific combinations of temperature and power. The 3D surface plot reveals that excessive heating or over-powering beyond the optimal line yields diminishing returns. Figure 13 demonstrates that higher temperatures reduce the power required per kilogram of CO produced. However, this does not account for the heat duty required to sustain high-temperature operation, which would need to be incorporated in future process-level design.

Product CO₂ Separation

As there is unreacted CO₂ in the CO product stream from the cathode of the electrolyzer, it has to undergo separation to provide the high purity CO product for the customers. In the literature, the two most common reported methods for CO₂ separation in the industry are pressure swing adsorption and amine absorption. Pressure swing adsorption works as a semi-batch process by having CO₂ being adsorbed on a solid sorbent at high pressure in a vessel, before being desorbed when the pressure is decreased [35]. An advantage of pressure swing adsorption is its relatively lower energy requirement than amine absorption [36]. However,

adsorption of CO₂ has a higher levelized cost at \$51/tonne CO₂ compared to amine absorption at \$49/tonne CO₂ [37]. Also, there is a relatively lower selectivity and capacity for CO₂, and this might not only lead to high impurities of CO₂ in the product stream, but also the loss of valuable product of CO [38]. Furthermore, pressure swing adsorption works at a relatively narrow range of 5-15 % v/v CO₂, and might not be able to offer the flexibility to accommodate a wide range of electrolyzer operation where the CO₂ conversion to CO is suboptimal [39].

On the other hand, amine absorption of CO₂ has been well established in the industry for post combustion carbon capture and sequestration (CCS) and offers relatively higher absorption efficiency of more than 90% [40]. Consisting of an absorber vessel and stripper vessel, amine absorption works where a CO₂ rich stream enters the bottom of the absorber with amine such as Monoethanolamine (MEA) entering from the top [41]. The amine then absorbs the CO₂ and leaves the absorber. This stream then enters the stripper where the amine solvent is regenerated and recycled back to the absorber, and the high purity CO₂ is captured. In the context of a SOEC for CO₂ to CO electrolysis as displayed in Figure 14, the gaseous stream leaving the top of the absorber will be high purity CO product, while the gaseous stream leaving the top of the stripper will be high purity CO₂ stream that can be recycled back to the electrolyzer input.

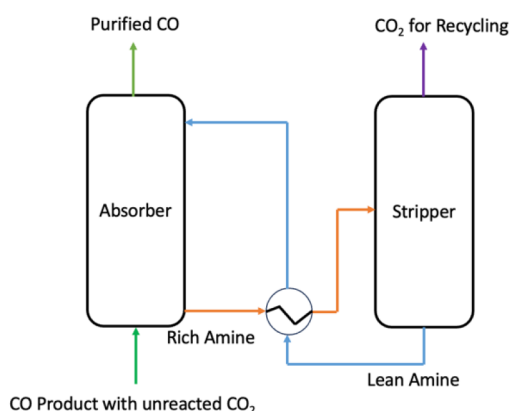


Figure 14—Amine absorption process for separation of unreacted CO₂ from SOEC product stream.

An advantage of amine absorption is having a continuous closed cycle operation where solvent replenishment is minimized, thus lowering operating costs. In addition, amines such as MEA has a natural affinity to CO₂ and operate at ambient pressure [42]. This lower operating pressure enables lower cost vessels. The main disadvantage of amine absorption for CO₂ separation is the high heating energy requirement to regenerate the solvent [41]. However, this energy demand can be met through heat integration with waste heat streams in chemical plants, thus saving operational costs. Also, due to the widespread utilization, existing or retired amine absorbers in chemical plants could be retrofitted and used to perform CO₂ separation, thus saving investment costs.

To investigate the effect of using an existing amine absorber to reduce CO₂ content of the SOEC CO product stream, we modify an Aspen Plus flue gas CO₂ removal example system as displayed in Figure 15. The composition of the flue gas entering the MEA absorber was changed to 77% CO with the remaining being CO₂, H₂O, and air, with an input flowrate of 72 kg/h. The purified CO leaving the absorber was able to recover 99.99% of the CO entering it, and this was expected as MEA has a high affinity to CO₂ but not CO. This example shows that it is possible to modify existing amine absorbers in chemical plants to separate the CO₂ from the SOEC CO product. This enables higher flexibility in deploying small scale SOEC units to provide CO feedstock.

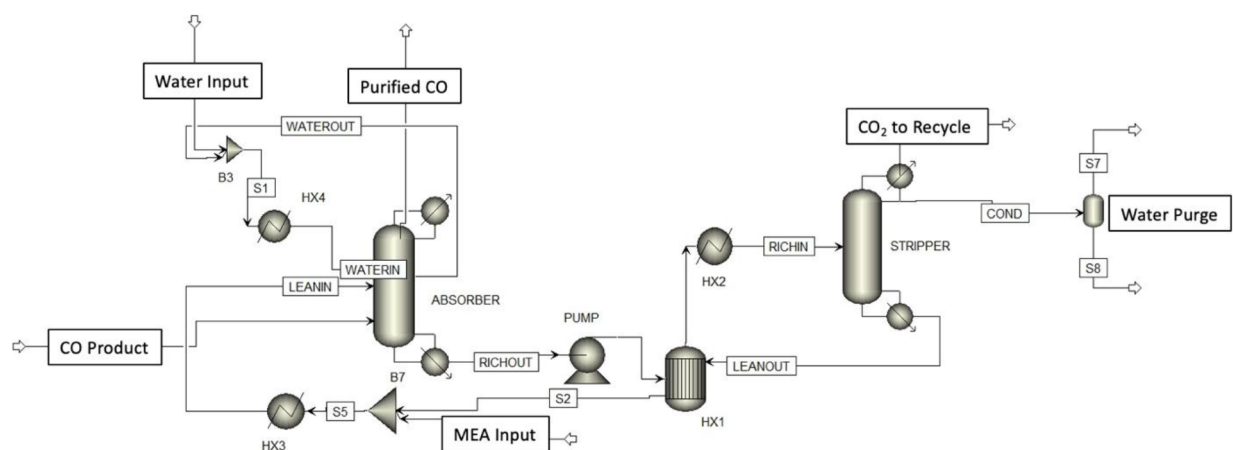


Figure 15—Modified Aspen Plus CO₂ removal system adapted for SOEC CO product stream.

Life Cycle Assessment (LCA)

LCA was performed using GREET to evaluate the total emissions of the SOEC resulting from the operations, including electricity and heating sources, and the production of the materials [43]. The system boundary for the LCA is the SOEC unit with the basis unit of kg CO_{2,eq} per kg CO produced. The sections of the LCA that were considered for the project are the SOEC materials production, stream losses, and the emissions for electricity and heating of the SOEC unit.

The electricity source was assumed to be natural gas-based electricity averaged in the US throughout the year. The required materials of the SOEC were Lanthanum Strontium Cobalt Oxide, 8%-Yttria-Stabilized Zirconia, Ceria, and Zirconia & Ni. These materials correspond to the anode, electrolyte, catalyst layer, and cathode, respectively. Also, there were some emissions of unreacted CO₂ from the SOEC are considered as stream losses. The heating requirements of the SOEC to operate at 700°C was assumed to be provided by a boiler, with parameters based on SOEC for H₂ production. These are the materials that will be considered for the LCA of the SOEC. There were some emissions of CO₂ that were not converted to CO and the amount of CO₂ in the outlet was also considered as a stream loss. Heating the SOEC to 700°C at startup and maintain the heat was considered for the SOEC. This was estimated from the hydrogen production SOEC systems as the system temperature and dimensions is fairly similar to SOEC production of CO.

The first step of the LCA is to find the total CO₂ equivalent emissions from the production of material, using both GREET inventory and literature values when required. The cell dimensions and the densities of each material were used to find the weight of the corresponding components of the SOEC. For the cathode material of Ni-YSZ, the emissions obtained from GREET inventory is 3.9 kg CO₂ per kg YSZ produced, and 10.9 kg CO₂ per kg of Ni produced. A literature suggested Ni to YSZ ratio of 30:70 was used to calculate the weight of the cathode [44]. The anode comprises of a thin layer of strontium with the rest being cobalt oxide and lanthanum with a weight ratio of 25:75 [45]. The emissions resulting from the strontium is assumed negligible due to its thin layer as well as the lack of data on its material construction. From GREET, the amount of emissions from the anode is then calculated using 33.3 kg CO₂ per kg lanthanum, 28.1 kg CO₂ per kg cobalt oxide, and 19.7 kg CO₂ per kg of ceria produced. The emissions of each material per tonne CO produced are displayed in Table 2.

Table 2—Material-specific emissions for components used in the SOEC stack (kg CO₂/tonne CO).

Material	Emissions (kg CO ₂ /tonne CO Produced)
Anode: Lanthanum	0.184
Anode: Cobalt Oxide	0.052
Anode: Ceria Doping Material	0.000
Electrolyte: 8%-Yttria-Stabilized Zirconia	0.005
Cathode: Ni	0.015
Cathode: YSZ	0.012

Apart from the materials, other emission sources from the SOEC are utilities such as heating and electricity. From the model, there is total of unreacted 0.34 kg CO₂ and a heating requirement of 2.73 kWh per kg CO produced. The emission of the heating was obtained by assuming 99.98 kg of CO₂ emitted per MMBtu of energy according to emission values of boilers [46] [47]. The SOEC also requires an electricity load of 4.35 kWh per kg CO produced, and the emissions of natural gas power reported in GREET is 0.2787 kg CO₂ per MJ. In addition, the SOEC has a net CO₂ consumption of 1.571 kg CO₂ per kg CO produced, which is counted as negative emission input. The emissions of the SOEC according to various sources are reported in Table 3. Although the process consumes CO₂ as a feedstock, the system results in net positive emissions due to high electricity and heating demands. This highlights the importance of using low-carbon or renewable energy sources to bring the overall footprint closer to net zero.

Table 3—Summary of total emissions from SOEC operation by source.

Component	Emissions (kg CO ₂ /tonne CO Produced)
CO ₂ Input	-1571
Heating	931
Electricity (Natural Gas)	2043
Material	0.27
Total	1402

Another comparison made in the LCA was to compare the amount of emission from each electricity source. The initial LCA was done assuming natural gas power as the source of electricity. compare the total emissions of the SOEC utilizing electricity from sustainable sources such as solar and wind energy, as well as nuclear energy. In the GREET inventory, the emission from solar and wind power is 0 kg CO₂ per MJ and the emission from nuclear power is 0.0016 kg CO₂ per MJ. Figure 16 displays the effect of using different power sources on the total emissions of the SOEC, where the renewable energy will result in a net negative emission.

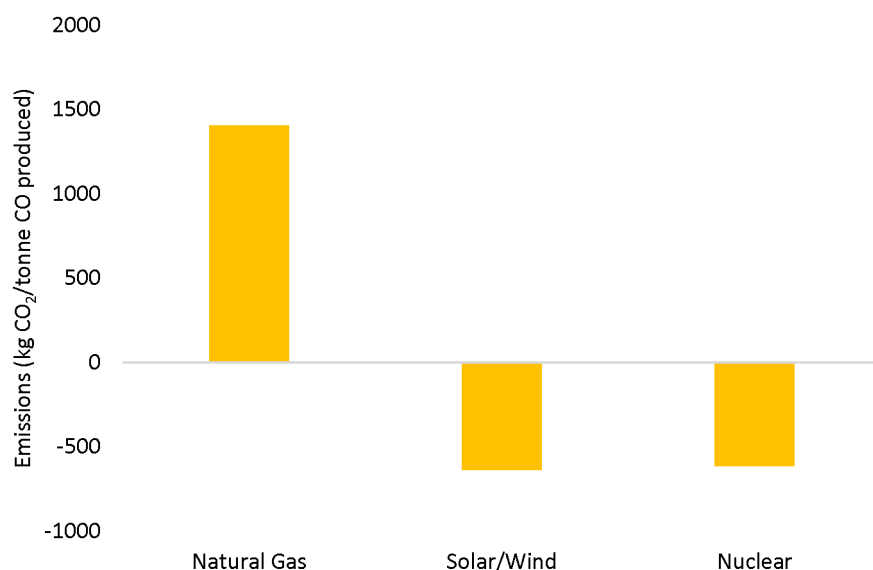


Figure 16—Comparison of total SOEC emissions based on different electricity sources.

Techno-economic Analysis (TEA)

A TEA was performed to determine the CAPEX and OPEX of the system. The calculation was done for both the SOEC unit and the CO₂ amine absorber. Cost values from SOEC H₂ electrolyzer from the literature was used to estimate the CAPEX of the SOEC for CO production, and the CAPEX of the CO₂ amine absorber was obtained from the Aspen Process Economic Analyzer in Aspen Plus. On the other hand, the OPEX is the cost of heating and electricity in both the SOEC and CO₂ amine absorber using utility costs in the literature. The results of the TEA are displayed in Table 4.

Table 4—Summary of SOEC and separation system CAPEX and OPEX (2023 USD per tonne of CO produced).

OPEX	SOEC	573
	Separator System	599
	Total	1172
CAPEX	SOEC	134
	Separator System	310
	Total	443
OPEX + CAPEX	SOEC	706
	Separator System	909
	Total	1615

One of the other aspects that was considered was the effect of the different source of electricity upon the over costs. The initial costing was done using natural gas, but then there was an investigation into wind and solar energy and nuclear energy. The costs are displayed in Figure 17. This showed that the costs of nuclear were the largest compared to the other costs, but it is considered a greener source of electricity as compared to natural gas. The other value was a combination of wind and solar energy as there is only a \$20/tonne CO produced difference than natural gas.

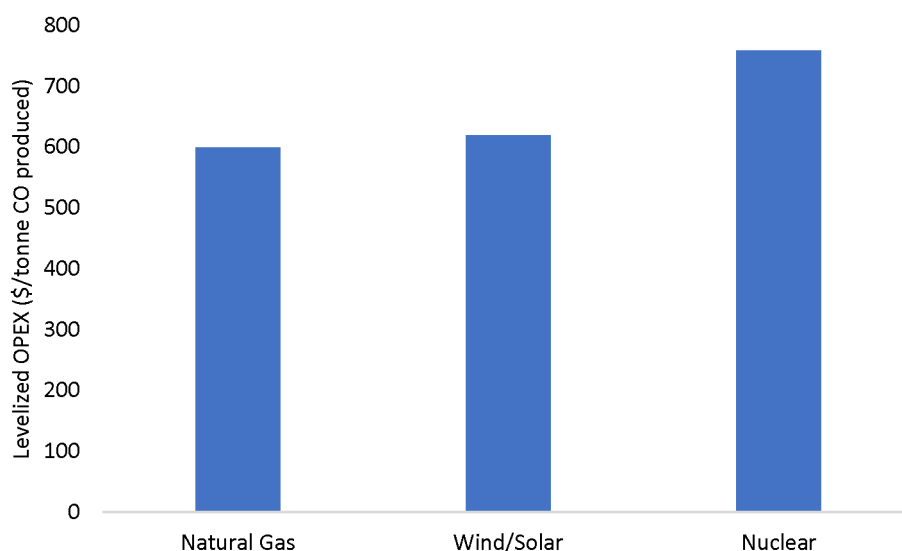


Figure 17—Impact of electricity source on total cost of CO production from SOEC.

Recommendations

The developed SOEC model in Aspen Custom Modeler incorporates key electrochemical reduction equations to simulate the conversion of CO₂ to CO, offering a valuable computational tool for evaluating the performance of CO₂ electrochemical reduction systems. However, the model currently relies on simplifying assumptions, including steady-state operation, isothermal and isobaric conditions, and the absence of oxygen back-diffusion from the anode to the cathode.

To enhance the model's fidelity and applicability to real-world systems, it is recommended to extend the current framework by incorporating transient behavior, non-isothermal and non-isobaric conditions, oxygen back-diffusion, detailed electrode kinetics, mass transport phenomena, material properties, gas flow dynamics, and catalyst effects. Specifically:

- **Transient Analysis:** Introduce time-dependent governing equations for species concentrations, temperature, and current density. This would enable simulation of system dynamics during startup, shutdown, and operational disturbances.
- **Non-Isothermal Modeling:** Include heat conduction equations to capture temperature gradients within the cell and assess the thermal behavior of the system.
- **Non-Isobaric Conditions:** Implement fluid dynamics models (e.g., Navier-Stokes equations) to simulate gas flow and pressure distribution throughout the cell.
- **Oxygen Back-Diffusion:** Integrate Fick's Law of diffusion to account for the transport of oxygen ions from the cathode to the anode, which could impact cell performance and efficiency.

Moreover, future model development could explore the production of alternative CO₂ electroreduction products beyond CO by incorporating reaction mechanisms and catalyst-specific kinetics.

In terms of the Life Cycle Assessment (LCA), further research is recommended to assess the environmental impact of SOEC material disposal. Currently, there is limited data on end-of-life treatment and environmental impacts associated with lanthanum-based perovskites and YSZ materials. As such information becomes available, it should be integrated into the LCA to enhance its completeness.

Additionally, the variability of renewable energy sources—such as intermittency of solar and wind power—should be considered in future assessments. Incorporating the energy storage systems required to ensure continuous operation, along with their associated costs and emissions, would provide a more realistic and comprehensive understanding of the overall system sustainability.

Conclusion

This study presents a comprehensive techno-economic analysis and life cycle assessment of CO₂ electrochemical conversion to CO using a Solid Oxide Electrolysis Cell (SOEC). A validated model was developed in Aspen Custom Modeler (ACM) to simulate SOEC performance under varying operational conditions. The model incorporated key electrochemical mechanisms and accurately predicted system behavior, showing good agreement with experimental data.

Sensitivity analysis revealed that temperature and applied power are the most critical factors influencing CO₂ conversion and energy efficiency. The integration of an amine absorption unit downstream of the SOEC provided a viable pathway to achieve high-purity CO by separating unreacted CO₂, with minimal loss of product and potential for process integration in existing chemical infrastructure.

The life cycle assessment showed that electricity sourcing plays a dominant role in determining the environmental impact of the system. When powered by renewable energy, the SOEC process achieves net-negative emissions, positioning it as a promising pathway for sustainable carbon utilization. The techno-economic analysis estimated a total cost of \$1,615/tonne of CO produced, with opportunities for cost reduction through improved heat integration and renewable electricity adoption.

Overall, this work demonstrates the feasibility of using SOEC technology for CO₂ conversion to CO in industrial applications and highlights key areas for future research and system optimization.

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